FIBRE PRODUCT CRITERION FOR RELATIVE LIMIT PRESERVATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1043-1081, following the statement at 1023-1041.

Editorial note. The sentence 'the proof is rather formal and we encourage the reader to find their own proof' is stylistic; the full argument is displayed.

Allowed standard prerequisites (editorial).

- Spaces over limits of presentations. lemma-space-over-limit (same chapter, cited at ~1052-1054). U = lim U_{i'} after base change to T.
- Directed limits with affine transitions exist. lemma-directed-inverse-system-has-limit (same chapter, card L26291, cited at ~1055-1056). U, U_{i'}, V, V_{i'} affine in the separated case.
- Characterization of locally finite presentation spaces. proposition-characterize-locally-finite-presentation (same chapter, card L26289, cited at ~1026-1027). Conditions (1) and (2) being extended by (3).

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SH: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 952–952. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 04AK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 694–694. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $X \to Y$ be a morphism of algebraic spaces over S. The equivalent conditions (1) and (2) of Proposition 04AK are also equivalent to*

(3) *for every directed limit $T = \lim T_i$ of quasi-compact and quasi-separated algebraic spaces T_i over S with affine transition morphisms the diagram of*

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sets

$$\begin{array}{ccc} \operatorname{colim}_i \operatorname{Mor}(T_i, X) & \longrightarrow & \operatorname{Mor}(T, X) \\ \downarrow & & \downarrow \\ \operatorname{colim}_i \operatorname{Mor}(T_i, Y) & \longrightarrow & \operatorname{Mor}(T, Y) \end{array}$$

is a fibre product diagram.

Proof. It is clear that (3) implies (2). We will assume (2) and prove (3). The proof is rather formal and we encourage the reader to find their own proof.

Let us first prove that (3) holds when T_i is in addition assumed separated for all i. Choose $i \in I$ and choose a surjective étale morphism $U_i \to T_i$ where U_i is affine. Using Lemma 07SH we see that with $U = U_i \times_{T_i} T$ and $U_{i'} = U_i \times_{T_i} T_{i'}$ we have $U = \lim_{i' \geq i} U_{i'}$. Of course U and $U_{i'}$ are affine (see Lemma 07SF). Since T_i is separated, the fibre product $V_i = U_i \times_{T_i} U_i$ is an affine scheme as well and we obtain affine schemes $V = V_i \times_{T_i} T$ and $V_{i'} = V_i \times_{T_i} T_{i'}$ with $V = \lim_{i' \geq i} V_{i'}$. Observe that $U \to T$ and $U_i \to T_i$ are surjective étale and that $V = U \times_T U$ and $V_{i'} = U_{i'} \times_{T_{i'}} U_{i'}$. Note that $\operatorname{Mor}(T, X)$ is the equalizer of the two maps $\operatorname{Mor}(U, X) \to \operatorname{Mor}(V, X)$; this is true for example because X as a sheaf on $(Sch/S)_{fppf}$ is the coequalizer of the two maps $h_V \to h_u$. Similarly $\operatorname{Mor}(T_{i'}, X)$ is the equalizer of the two maps $\operatorname{Mor}(U_{i'}, X) \to \operatorname{Mor}(V_{i'}, X)$. And of course the same thing is true with X replaced with Y. Condition (2) says that the diagrams of in (3) are fibre products in the case of $U = \lim U_i$ and $V = \lim V_i$. It follows formally that the same thing is true for $T = \lim T_i$.

In the general case, choose an affine scheme U, an $i \in I$, and a surjective étale morphism $U \to T_i$. Repeating the argument of the previous paragraph we still achieve the proof: the schemes $V_{i'}$, V are no longer affine, but they are still quasi-compact and separated and the result of the preceding paragraph applies. □